

VIBE ROUNDS

Clinical Decision Support System (CDSS) Mode

Concept & Architecture Document

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Executive Summary

DEFINING STATEMENT

CDSS Mode = A real-time skeptic auditor performing Socratic enquiry, flagging critical reasoning gaps as they occur at the bedside.

Vibe Rounds CDSS Mode is the third and most clinically consequential domain of the Vibe Rounds framework. Where Domain 1 (clinical education) and Domain 2 (guided discovery research) operate in learning and analytical contexts, CDSS Mode enters the active care environment — functioning as a cognitive safety layer that runs alongside clinical decision-making in real time.

This document defines the concept, architecture, clinical rationale, and implementation vision for CDSS Mode.

The Problem: Why Bedside Reasoning Fails

Clinical Reasoning Errors Are Not Knowledge Gaps

The majority of preventable clinical errors do not arise from a lack of medical knowledge. Research in cognitive medicine consistently shows that most diagnostic failures stem from:

- Premature closure — settling on a diagnosis before adequate information has been gathered
- Anchoring bias — over-weighting the first piece of clinical information received
- Missed differentials — failing to consider an alternative diagnosis that could explain the findings

- Context loss — losing track of the patient's timeline, comorbidities, or prior investigations
- Cross-specialty blind spots — not recognising when a case crosses into another discipline
- Logic jumps — advancing to management decisions without completing the diagnostic reasoning chain

These errors are particularly prevalent under high cognitive load conditions — end of long shifts, overnight hours, complex multi-system presentations, and crowded clinical environments. They affect trainees and experienced consultants alike.

INSIGHT

The solution is not more information. It is structured, real-time challenge of the reasoning process itself.

CDSS Mode: Concept Definition

What It Is

Vibe Rounds CDSS Mode is a real-time clinical reasoning audit engine embedded within the care environment. It is not a diagnostic tool, a reference system, or a clinical guideline navigator. It is a cognitive safety layer — an AI-powered Socratic auditor that monitors clinical reasoning as it unfolds and surfaces structured challenges at critical decision points.

CORE IDENTITY

Not a tutor. Not a coder. Not an answer engine. A cognitive safety and reasoning quality monitoring system — active at the bedside, in real time.

What It Is Not

- Not a diagnostic oracle that provides answers
- Not a replacement for clinical judgment
- Not a passive reference database
- Not a documentation tool in primary function
- Not a protocol navigator or guideline checker

The CDSS Mode Paradigm Shift

Traditional CDSS tools are reactive: a clinician enters data, the system returns outputs. Vibe Rounds CDSS Mode is bidirectional and Socratic — it asks back. The fundamental shift is from:

Traditional CDSS	Vibe Rounds CDSS Mode
Clinician asks → AI answers	AI also questions → Clinician thinks
Reference-based	Reasoning-based
Point-in-time lookup	Continuous reasoning audit
Passive alerts	Active Socratic challenge
Knowledge delivery	Cognitive quality monitoring

Core Functions of CDSS Mode

When activated, CDSS Mode performs the following seven audit functions continuously across the clinical encounter:

#	Function	Description
1	Differential Flag	Identifies when critical alternative diagnoses are absent from the clinician's stated reasoning — e.g. 'You've prioritised pneumonia; have you excluded PE and cardiac failure?'
2	Premature Closure Detection	Detects when a management plan is advancing before the differential has been adequately narrowed — challenges commitment before evidence is sufficient.
3	Context Gap Highlighting	Flags missing contextual elements: timeline inconsistencies, undocumented red flags, unaddressed comorbidities, or absent family/social history relevant to the presentation.
4	Cross-Specialty Blind Spot Check	Identifies when a presentation pattern suggests involvement of a specialty outside the primary team's domain and prompts appropriate consideration.
5	Logic Jump Detection	Surfaces gaps in the reasoning chain where the clinician has moved from symptom to diagnosis or diagnosis to treatment without documenting the inferential steps.
6	Targeted Socratic Questions	Delivers precise, evidence-informed Socratic challenges — 'Why this diagnosis?', 'What is the single finding that would make you change this plan?', 'What could kill this patient that you haven't yet excluded?'
7	Reasoning Quality Audit	Generates a structured reasoning quality report on demand: differentials considered, bias risks flagged, decision rationale captured — for learning, audit, and medico-legal documentation.

The Swiss Knife Framework

CDSS Mode as a Tool Layer, Not a Standalone System

CDSS Mode is not designed to replace existing clinical infrastructure. It is conceived as an additional layer in the Vibe Rounds 'Swiss knife' — a modular, activatable tool that sits alongside other capabilities:

User Role	Student / Clinical Analyst
AI Role	Active Socratic Auditor + Polymath Guide
Expert Network	Peer and Supervisor Connection Layer
Activation Model	Mode-based — clinician enables CDSS Mode on demand
Interaction Style	Bidirectional: AI asks as much as it answers
Prompts Directory	Community-curated LLM prompt library for clinical contexts

The Paradigm Addition: AI That Asks

The core philosophical innovation of CDSS Mode — and Vibe Rounds overall — is the inversion of the default AI interaction model. Standard AI systems wait to be asked. CDSS Mode activates a paradigm where:

THE PARADIGM SHIFT

Rather than only answering what the user asks, CDSS Mode enables a paradigm where the AI asks and the clinician thinks.

This is not a passive feature. It is a deliberate cognitive scaffolding mechanism grounded in decades of educational psychology research on Socratic method, metacognition, and reflective practice in professional domains.

The Learning Dimension: Why CDSS Mode is Also Education

Passive vs Active Clinical Intelligence

Consider two approaches to a diabetes management case:

Passive Approach	Study the diabetes guideline, then ask AI to help apply it to the patient.
Active Approach	Design the management plan yourself first, then ask AI to evaluate and improve it.

Optimal Approach	Design the plan yourself — particularly if building competency — then use AI's Socratic feedback to identify gaps and refine reasoning. The clinician remains the decision-maker; the AI is the auditor.
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The optimal approach — which CDSS Mode operationalises — is not only more educationally effective. It is also less clinically risky. It keeps the human in the reasoning loop, builds genuine competency over time, and provides a 24/7 clinical reasoning AI plus self-audit capability.

SAFETY RATIONALE <i>CDSS Mode is not a shortcut to answers. It is a 24×7 clinical reasoning AI and self-audit engine — the safest and most effective configuration for AI in active clinical care.</i>
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System Architecture: Vision

Core Components

- **Voice and text input at the bedside — no form-filling, no menu navigation** Natural Language Interface
- **Structured logic layer that maps clinical inputs against standard reasoning frameworks and flags deviations** Reasoning Audit Engine
- **Context-aware question generation calibrated to the clinical scenario, the clinician's stated reasoning, and known bias patterns** Socratic Question Generator
- **A community-curated library of LLM prompt templates designed by clinical experts — accessible, scrollable, shareable, and modifiable** Prompts Directory
- **FHIR-compatible data ingestion for real-time patient context (timeline, investigations, medications, prior diagnoses)** EMR Integration Layer
- **Image analysis (wounds, ECGs, imaging), audio transcription, and structured data from connected devices** Multimodal Input Processing
- **On-demand structured output: differentials considered, biases flagged, reasoning chain documented** Reasoning Quality Report Generator

CDSS Mode Activation Model

CDSS Mode is not always-on by default. It is activated by the clinician as a deliberate choice to enter a structured reasoning audit session. This preserves clinical autonomy and prevents cognitive overload from unsolicited interruptions.

Trigger	Clinician activates CDSS Mode via voice or tap command
Input	Clinical narrative, patient context, stated reasoning or plan
Processing	Reasoning audit engine analyses for the seven core functions

Output	Targeted Socratic questions and/or flagged gaps — not answers
Decision	Clinician retains full decision-making authority
Log	Optional session log for audit, reflection, and quality improvement

The Prompts Directory

Democratising Clinical AI Prompting

A key component of the Vibe Rounds Swiss knife is the Prompts Directory: a scrollable, searchable library of LLM prompt templates designed by clinicians, educators, and domain experts. This addresses a fundamental access problem:

EQUITY RATIONALE

Prompt engineering in healthcare should not be a skill reserved for those with technical backgrounds. The Prompts Directory brings expert-designed cognitive models to every clinician, at every level, in every setting.

Directory Features

- Browse and filter by specialty, clinical context, reasoning type, or difficulty level
- Peer-reviewed and community-rated prompts
- Modifiable templates — adapt to local context, patient complexity, or teaching level
- Contributed by clinicians, academics, and educators — not only AI engineers
- Integrated with CDSS Mode — select a prompt as the audit framework for a specific case type

Safety, Ethics, and Scope

What CDSS Mode Does Not Do

- Does not make diagnoses or recommend treatments as a standalone system
- Does not override clinical judgment — the clinician is always the decision-maker
- Does not access patient data without institutional consent and infrastructure
- Does not replace specialist consultation
- Does not function as a medico-legal substitute for documented clinical assessment

Appropriate Use Boundaries

CDSS Mode is designed for use within an institutional framework that provides:

- Appropriate data governance and patient consent infrastructure
- Clinical oversight and validation of AI outputs before deployment
- Training for clinicians in the appropriate use of AI-augmented reasoning
- Integration with existing clinical governance and quality improvement processes

DISCLAIMER

CDSS Mode is a concept and early-stage architecture. It is not currently validated for clinical deployment. Institutional and regulatory validation is required before any clinical use.

Implementation Roadmap

Stage 1 — Concept	Architecture defined. Core philosophy and function set documented. Vision paper published. (Current stage — June 2026)
Stage 2 — Prototype	Prompt-based CDSS Mode demonstrated in controlled simulation environments using existing LLM infrastructure (no EMR integration).
Stage 3 — Pilot	Structured pilot in a clinical education setting — residents or interns using CDSS Mode for case preparation and post-round reflection.
Stage 4 — EMR Integration	FHIR-compatible integration with a partner EMR for real-time patient context ingestion in a controlled institutional environment.
Stage 5 — Validation	Multi-site clinical validation: reasoning quality metrics, error detection rates, user experience, and safety outcomes.
Stage 6 — Deployment	Institutional deployment with governance framework, training programme, and continuous quality monitoring.

Conclusion

Vibe Rounds CDSS Mode represents the most ambitious and highest-consequence application of the Vibe Rounds philosophy. Where educational applications support learning and research applications support discovery, CDSS Mode enters the clinical moment itself — the space where errors have immediate patient consequences.

Its design is grounded in a clear philosophical commitment: AI in clinical care should augment the clinician's reasoning, not replace it. By functioning as a real-time Socratic auditor rather than an answer engine, CDSS Mode is positioned to improve diagnostic quality, reduce cognitive error, and build genuine clinical competency — at any hour, in any setting, for any clinician willing to think with it.

VISION

The goal of Vibe Rounds CDSS Mode is the best version of every clinician's reasoning — at every hour, for every patient.

How to Cite

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